$$\begin{aligned}
\cos 3x &= \cos(2x + x) \\
&= \cos 2x \cos x - \sin 2x \sin x \\
&= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x \\
&= \boxed{4\cos^3 x - 3\cos x} \\
\sin 3x &= \sin(2x + x) \\
&= \sin 2x \cos x + \cos 2x \sin x \\
&= 2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x \\
&= \boxed{3\sin x - 4\sin^3 x}
\end{aligned}$$

(i)

$$\cos 3\alpha = 4c^3 - 3c$$

$$\begin{aligned}
I(\alpha) &= \int_0^{\alpha} (\sin x + 2\sin 3x)\, dx \\
&= \left[-\cos x - \frac{2}{3}\cos 3x \right]_0^{\alpha} \\
&= -\cos\alpha - \frac{2}{3}\cos 3\alpha + \frac{5}{3} \\
&= -c - \frac{2}{3}(4c^3 - 3c) + \frac{5}{3} \\
&= \boxed{-\frac{8}{3}c^3 + c + \frac{5}{3}}
\end{aligned}$$

When $c = 1$, $\quad I(\alpha) = -\dfrac{8}{3} + 1 + \dfrac{5}{3} = 0$

(ii)

Let $c = \cos\beta$. Using Eustace's method,

$$\begin{aligned}
I'(\beta) &= \left[\frac{7}{2}\sin^2 x - 2\sin^4 x \right]_0^{\beta} \\
&= \frac{7}{2}\sin^2\beta - 2\sin^4\beta \\
&= \frac{7}{2}(1 - c^2) - 2(1 - c^2)^2 \\
&= \frac{3}{2} + \frac{1}{2}c^2 - 2c^4
\end{aligned}$$

When $c = -\dfrac{1}{6}$.

$$I(\beta) = -\frac{8}{3}\left(-\frac{1}{6}\right)^3 - \frac{1}{6} + \frac{5}{3} = \frac{245}{162}$$

$$I'(\beta) = \frac{3}{2} + \frac{1}{2}\left(-\frac{1}{6}\right)^2 - 2\left(-\frac{1}{6}\right)^4 = \frac{245}{162}$$

$$\boxed{I'(\beta) = I(\beta)}$$

When $\frac{3}{2} + \frac{1}{2}c^2 - 2c^4 = -\frac{8}{3}c^3 + c + \frac{5}{3}$

$$(c-1)^2(2c+1)(6c+1) = 0$$

$$c = 1, \qquad c = -\frac{1}{2}, \qquad c = -\frac{1}{6}$$

Since $c = \cos\alpha$

$$\boxed{\alpha = 2k\pi, \qquad \alpha = 2k\pi \pm \frac{2\pi}{3}, \qquad \alpha = 2k\pi \pm \arccos\left(-\frac{1}{6}\right)}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 13, 2026